Git Commands Quick Reference

Setup and Config

Command	Description
`git config --global user.name "name"`	Set the name attached to commits
`git config --global user.email "email"`	Set the email attached to commits
`git config --global color.ui auto`	Enable colorization of command output
`git config --list`	List all configured settings
`git help <command>`	Get help for a Git command

Sharing and Updating Projects

Command	Description
`git remote -v`	List all configured remotes
`git remote add <name> <url>`	Add a new remote
`git remote remove <name>`	Remove a remote
`git fetch`	Download objects and refs from remote
`git fetch <remote>`	Fetch from specific remote
`git pull`	Fetch and merge from remote
`git pull --rebase`	Fetch and rebase from remote
`git push`	Push changes to remote
`git push <remote> <branch>`	Push branch to remote
`git push -u <remote> <branch>`	Push and set upstream
`git push --force-with-lease`	Safely force push
`git push --tags`	Push all tags to remote

Patching

Command	Description
`git cherry-pick <commit>`	Apply a commit to current branch
`git cherry-pick --abort`	Abort cherry-pick
`git rebase <branch>`	Rebase current branch onto another
`git rebase -i HEAD~<n>`	Interactive rebase last n commits
`git rebase --continue`	Continue after resolving conflicts
`git rebase --abort`	Abort rebase
`git revert <commit>`	Create new commit that undoes a commit
`git revert --no-commit <commit>`	Revert without auto-committing

Getting and Creating Projects

Command	Description
`git init`	Initialize a new Git repository
`git init <directory>`	Create a new local repository in specified directory
`git clone <url>`	Clone a repository from a remote source
`git clone <url> <directory>`	Clone a repository into a specific directory

Branching and Merging

Command	Description
`git branch`	List all local branches
`git branch -a`	List all branches (local and remote)
`git branch <branch>`	Create a new branch
`git branch -d <branch>`	Delete a branch
`git branch -m <new-name>`	Rename current branch
`git checkout <branch>`	Switch to a branch
`git checkout -b <branch>`	Create and switch to new branch
`git switch <branch>`	Switch to a branch (modern)
`git switch -c <branch>`	Create and switch to new branch (modern)
`git merge <branch>`	Merge a branch into current branch
`git merge --no-ff <branch>`	Merge with a merge commit
`git merge --abort`	Abort a merge in progress
`git log`	Show commit history
`git log --oneline`	Show compact commit history
`git log --graph --all`	Show branch graph
`git stash`	Temporarily save uncommitted changes
`git stash pop`	Apply and remove most recent stash
`git stash list`	List all stashes
`git tag <name>`	Create a lightweight tag
`git tag -a <name> -m "msg"`	Create an annotated tag

Debugging

Command	Description
`git bisect start`	Start binary search for bad commit
`git bisect bad`	Mark current commit as bad
`git bisect good <commit>`	Mark commit as good
`git bisect reset`	End bisect session
`git blame <file>`	Show who changed each line
`git blame -L 10,20 <file>`	Blame specific line range
`git grep <pattern>`	Search for pattern in tracked files

Basic Snapshotting

Command	Description
`git status`	Show the status of the working directory
`git add <file>`	Add a file to the staging area
`git add .`	Add all new and changed files to staging
`git add -p`	Interactive staging - choose hunks to add
`git diff`	Show unstaged changes
`git diff --staged`	Show staged changes
`git commit -m "message"`	Commit staged changes with a message
`git commit -am "message"`	Stage all tracked files and commit
`git commit --amend`	Amend the last commit
`git restore <file>`	Discard changes in working directory
`git restore --staged <file>`	Unstage a file
`git reset <file>`	Unstage a file (keep changes)
`git reset --hard HEAD`	Discard all local changes
`git rm <file>`	Remove file from working directory and staging
`git mv <old> <new>`	Rename or move a file

Inspection and Comparison

Command	Description
`git show <commit>`	Show changes of a commit
`git show HEAD`	Show last commit
`git log`	Show commit history
`git log --follow <file>`	Show commit history for file
`git log --author="name"`	Filter commits by author
`git log --grep="pattern"`	Search commit messages
`git log --since="2 weeks ago"`	Filter by date
`git diff <branch1>..<branch2>`	Compare two branches
`git diff <commit1> <commit2>`	Compare two commits
`git shortlog`	Summarized commit history

Administration

Command	Description
`git clean -n`	Show what would be deleted
`git clean -f`	Delete untracked files
`git clean -fd`	Delete untracked files and directories
`git gc`	Cleanup and optimize local repository
`git fsck`	Verify database integrity
`git reflog`	Show reference log (recovery tool)
`git reflog show <branch>`	Show reflog for specific branch